# CONNERS CPT 3™

## Continuous Performance Test 3rd Edition™

C. Keith Conners, Ph.D.

# Assessment Report

| | |
|---|---|
| **Name/ID:** | Kyle Rosebrook |
| Age: | 36 |
| Gender: | Male |
| Birth Date: | January 16, 1986 |
| Administration Date: | July 7, 2022 |
| Normative Option: | Gender Specific norms |
| Input Device: | Keyboard |
| Assessor's Name: | |
| Medication/Notes: | |

This Assessment Report is intended for use by qualified assessors only, and is not to be shown or presented to the respondent or any other unqualified individuals or used as the sole basis for clinical diagnosis or intervention. Administrators are cautioned against drawing unsupported interpretations. To obtain a comprehensive view of the individual, information from this report should be combined with information gathered from other psychometric measures, interviews, observations, and available records. This report is based on an algorithm that produces the most common interpretations of the obtained scores. Additional interpretive information is found in the *Conners CPT 3 Manual* (published by MHS).

MHS

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# Introduction

The Conners Continuous Performance Test 3rd Edition (Conners CPT 3™) assesses attention-related problems in individuals aged 8 years and older. During the 14-minute, 360-trial administration, respondents are required to respond when any letter appears, except the non-target letter "X." By indexing the respondent's performance in areas of inattentiveness, impulsivity, sustained attention, and vigilance, the Conners CPT 3 can be a useful adjunct to the process of diagnosing Attention-Deficit/Hyperactivity Disorder (ADHD), as well as other psychological and neurological conditions related to attention.

# Validity of Administration

The Conners CPT 3 performs a validity check based on the number of hits and omission errors committed, as well as a self-diagnostic check of the accuracy of the timing of each administration. If there is an insufficient number of hits to compute scores, and/or if the omission error rate exceeds 25%, these issues will be noted. Also, the program will issue a warning message noting that the administration was invalid if a timing issue is detected.

There was no indication of any validity issues; the current administration should be considered valid.

# Response Style Analysis

The variable ***C*** represents an individual's natural response style in tasks that involve a speed-accuracy trade-off. Based on his or her score on this variable, a respondent can be classified as having one of the following three response styles: a **conservative** style (T-score ≥ 60) of responding that emphasizes accuracy over speed; a **liberal** style (T-score ≤ 40) of responding that emphasizes speed over accuracy; or a **balanced** style (T-score = 41-59) of responding that is sensitive to both speed and accuracy. Based on Kyle's responses, **he has a balanced style of responding that is sensitive to both speed and accuracy (T-score = 41)**. This response style is not likely to bias other Conners CPT 3 scores.

# T-score Guidelines

The guidelines in the following table apply to all T-scores in this report.

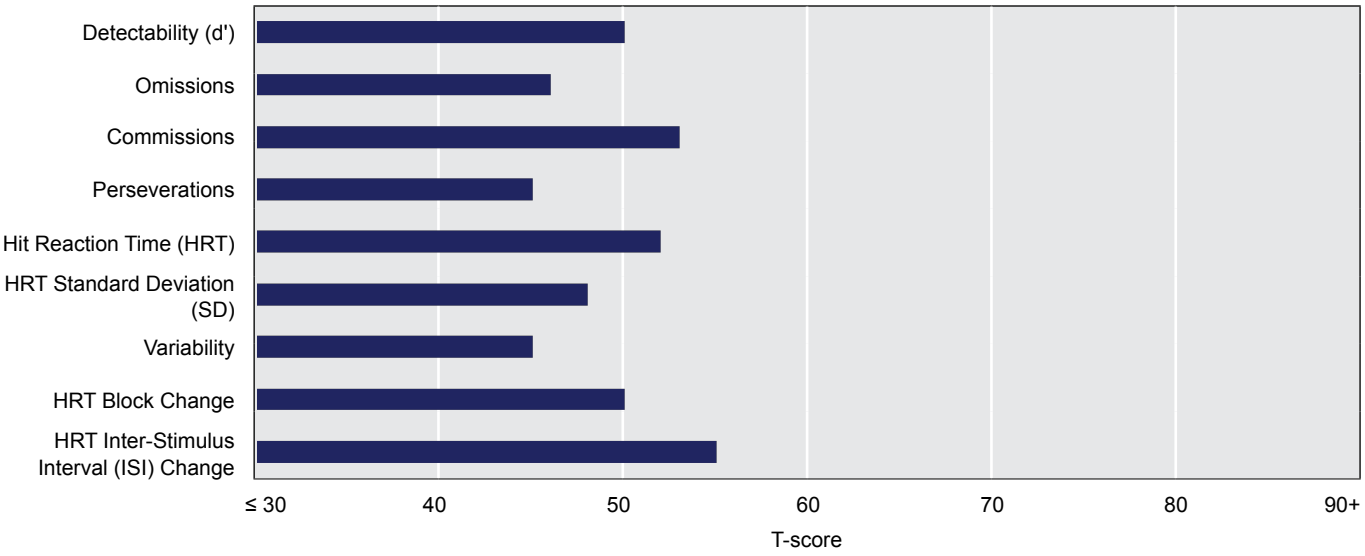
| Guidelines | | | |
|---|---|---|---|
| T-score | For Hit Reaction Time (HRT) | T-score | For all other variables |
| 70+ | **Atypically Slow** | 70+ | **Very Elevated** |
| 60-69 | **Slow** | 60-69 | **Elevated** |
| 55-59 | A Little Slow | 55-59 | High Average |
| 45-54 | Average | 45-54 | Average |
| 40-44 | **A Little Fast** | < 45 | Low |
| < 40 | **Atypically Fast** | | |


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# Overview of Conners CPT 3 Scores

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This section provides an overview of Kyle's Conners CPT 3 scores.




| Variable Type | Measure | T-score | Guideline | Interpretation |
|---|---|---|---|---|
| Detectability | d' | 50 | Average | **Average** ability to differentiate targets from non-targets. |
| Error Type | Omissions | 46 | Average | **Average** rate of missed targets. |
| | Commissions | 53 | Average | **Average** rate of incorrect responses to non-targets. |
| | Perseverations | 45 | Average | **Average** rate of random, repetitive, or anticipatory responses. |
| Reaction Time Statistics | HRT | 52 | Average | **Average** mean response speed. |
| | HRT SD | 48 | Average | **Average consistency** in reaction times. |
| | Variability | 45 | Average | **Average** variability in reaction time consistency. |
| | HRT Block Change | 50 | Average | **Average** change in response speed in later blocks. |
| | HRT ISI Change | 55 | High Average | **Slight** reduction in response speed at longer ISIs. |

**Summary:** None of Kyle's T-scores were atypical.

Overall, the results do not suggest that Kyle has a disorder characterized by attention deficits, such as ADHD.

Although there is no pervasive pattern of atypical T-scores, Kyle's response pattern indicate a possible issue with one or more specific dimensions of attention:

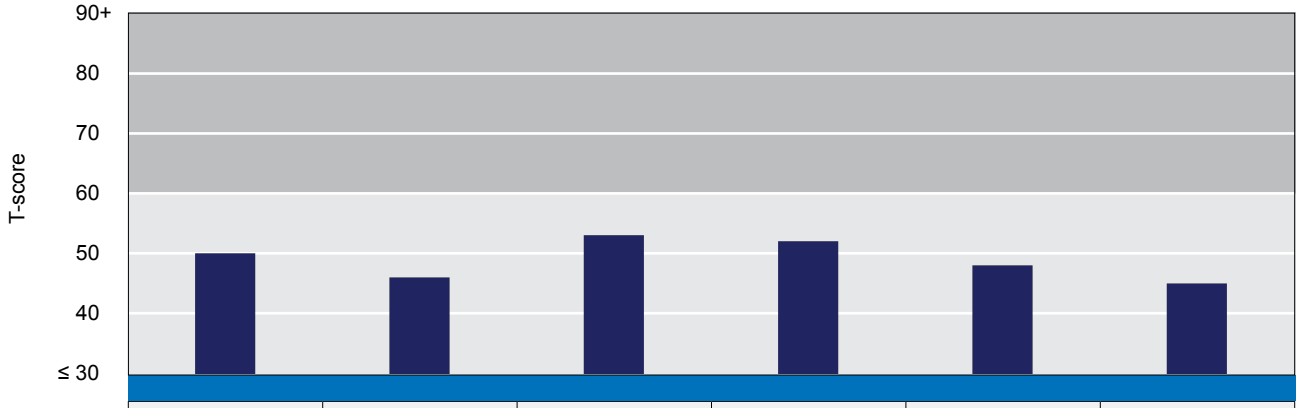
- **Sustained Attention (Some Indication)**


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# Measures of Inattentiveness

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This section summarizes Kyle's scores on the inattentiveness measures and provides information about how he compares to the normative group. Indicators of inattentiveness on the Conners CPT 3 are poor Detectability (d'), a high percentage of Omissions and Commissions, a slow Hit Reaction Time (HRT), as well as high levels of inconsistency in response speed (Hit Reaction Time Standard Deviation [HRT SD] and Variability).




| | Detectability (d') | Omissions | Commissions | HRT | HRT SD | Variability |
|---|---|---|---|---|---|---|
| T-score | 50 | 46 | 53 | 52 | 48 | 45 |
| Guideline | Average | Average | Average | Average | Average | Average |

**Detectability (d')** measures the respondent's ability to differentiate non-targets (i.e., the letter X) from targets (i.e., all other letters). Kyle's T-score is 50 and falls in the **Average** range. This result means that his ability to discriminate non-targets from targets was average when compared to the normative group.

**Omissions** result from a failure to respond to targets. Kyle's T-score is 46 and falls in the **Average** range. This result means that he missed an average percentage of targets when compared to the normative group.

**Commissions** are made when responses are given to non-targets. Kyle's T-score is 53 and falls in the **Average** range. This result means that he responded to an average percentage of non-targets when compared to the normative group.

**HRT** is the mean response speed of correct responses for the whole administration. Kyle's T-score is 52 and falls in the **Average** range. This result means that his response speed was about the same as the normative group's response speed.

**HRT SD** is a measure of response speed consistency during the entire administration. Kyle's T-score is 48 and falls in the **Average** range. This result means that his response speed was about as consistent as the normative group.

**Variability**, like HRT SD, is a measure of response speed consistency; however, Variability is a "within respondent" measure; that is, the amount of variability that Kyle showed in 18 separate segments of the administration in relation to his own overall HRT SD. Kyle's T-score is 45 and falls in the **Average** range. This result means his response speed variability was average when compared to the normative group.

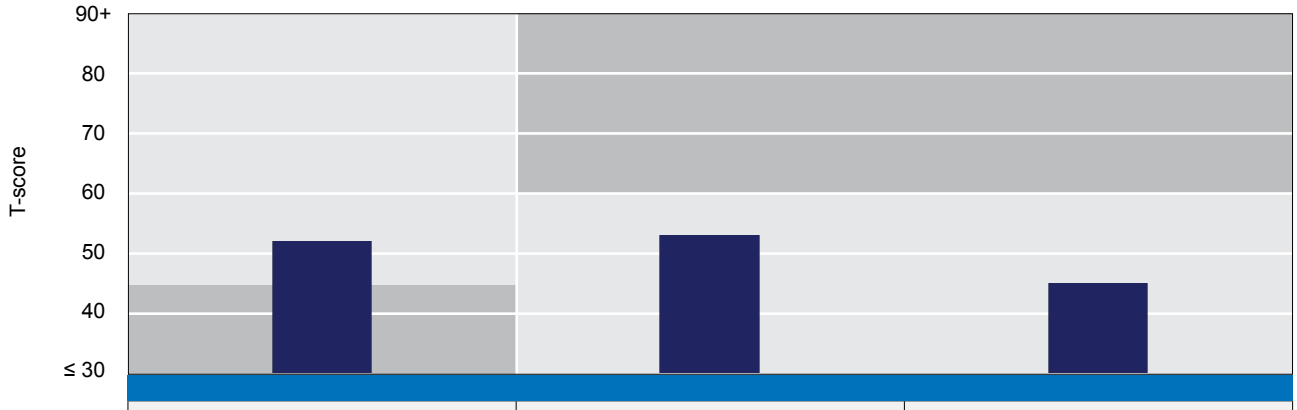
**Kyle's scores on these measures do not indicate a problem with inattentiveness.**


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# Measures of Impulsivity

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This section summarizes Kyle's scores on the impulsivity measures and provides information about how he compares to the normative group. Indicators of impulsivity on the Conners CPT 3 include a faster than normal Hit Reaction Time (HRT) in addition to a higher than average rate of Commissions and/or Perseverations.




| | HRT | Commissions | Perseverations |
|---|---|---|---|
| T-score | 52 | 53 | 45 |
| Guideline | Average | Average | Average |

**HRT** is the mean response speed of correct responses for the whole administration. Kyle's T-score is 52 and falls in the **Average** range. This result means that his response speed was about the same as the normative group's response speed.

**Commissions** are made when responses are given to non-targets. Kyle's T-score is 53 and falls in the **Average** range. This result means that he responded to an average percentage of non-targets when compared to the normative group.

**Perseverations** are random or anticipatory responses. Kyle's T-score is 45 and falls in the **Average** range. This result means that he made approximately the same number of perseverative errors when compared to the normative group.

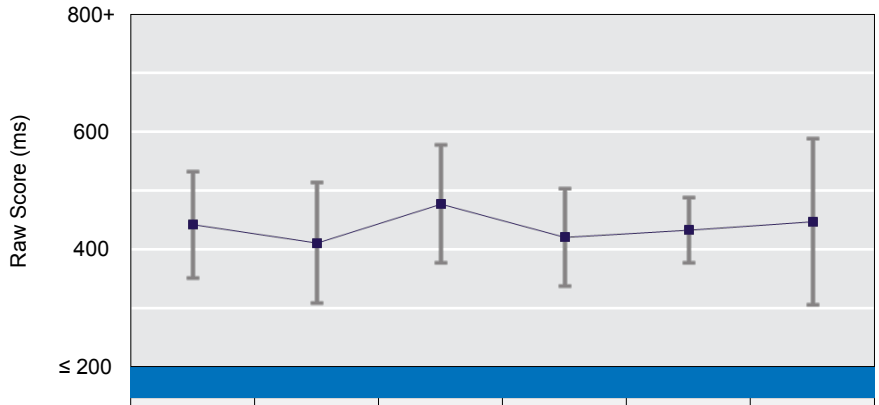
**Kyle's scores on these measures do not indicate a problem with impulsivity.**


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# Measures of Sustained Attention

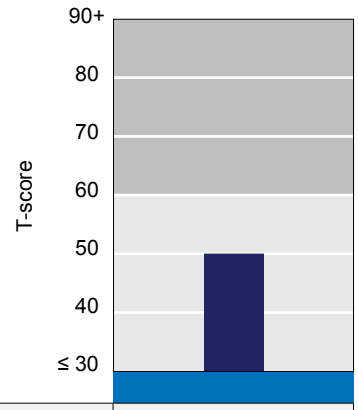
This section summarizes Kyle's scores on the sustained attention measures. Sustained attention is defined as the respondent's ability to maintain attention as the administration progresses. A decrease in sustained attention across time is captured by atypical slowing in the respondent's Hit Reaction Times (HRT; as indicated by the variable HRT Block Change), as well as by increases in Omissions and Commissions in later blocks of the administration.

## Hit Reaction Time by Block




| | Block 1 | Block 2 | Block 3 | Block 4 | Block 5 | Block 6 |
|---|---|---|---|---|---|---|
| HRT (ms) | 442 | 410 | 477 | 420 | 432 | 447 |
| HRT SD (ms) | 91 | 103 | 101 | 84 | 55 | 141 |

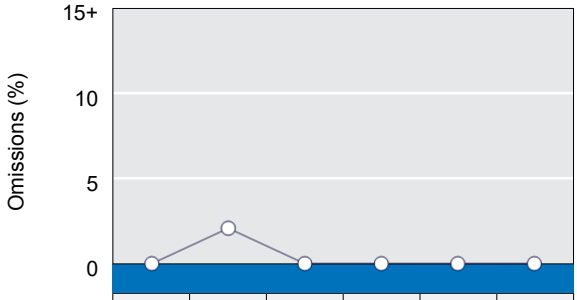
## HRT Block Change




| | |
|---|---|
| T-score | 50 |
| Guideline | Average |

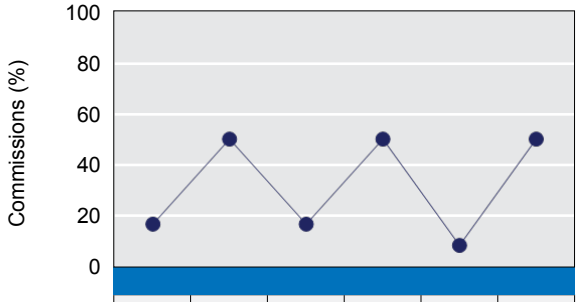
**Note.** ms = milliseconds; SD = Standard Deviation.

## Omissions by Block




| | Block 1 | Block 2 | Block 3 | Block 4 | Block 5 | Block 6 |
|---|---|---|---|---|---|---|
| Omissions (%) | 0 | 2 | 0 | 0 | 0 | 0 |

## Commissions by Block




| | Block 1 | | Block 2 | | Block 3 | | Block 4 | | Block 5 | | Block 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Commissions (%) | 17 | < | 50 | < | 17 | < | 50 | < | 8 | < | 50 * |

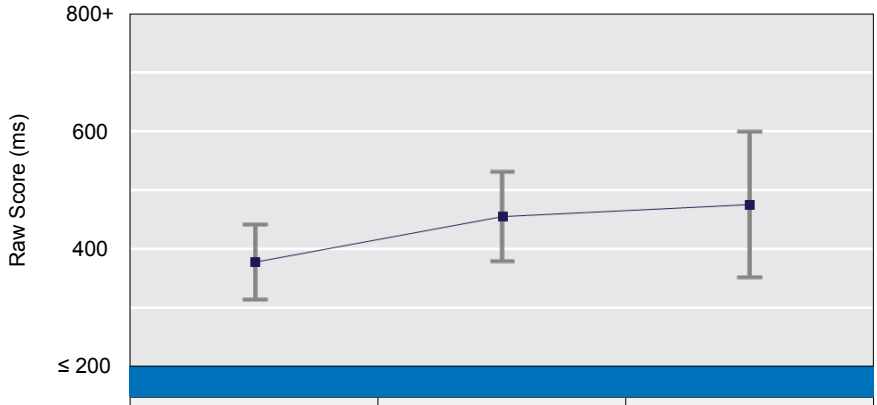
**Note**. The < symbol indicates that the error rate of the later block is significantly ($p < .10$) higher than the error rate of the previous block. The * symbol indicates that the error rate in Block 6 is significantly ($p < .10$) higher than the error rate in Block 1.

**HRT Block Change** indicates the change in mean response speed across blocks. Kyle's T-score is 50 and falls in the **Average** range. This result means that he had an average reduction in response speed in later blocks. In terms of error rates, Kyle's commission errors increased significantly ($p < .10$) across blocks, but his omission errors did not signficantly increase across multiple adjacent blocks. **Kyle's profile of scores on these measures indicates some support for a problem with sustained attention.**


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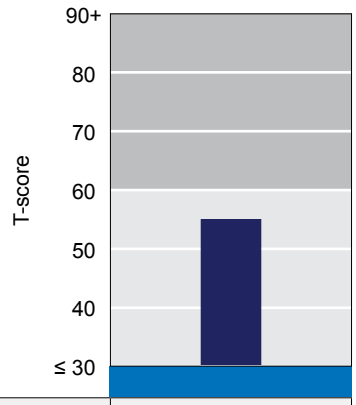
This section summarizes Kyle's scores on the vigilance measures. Vigilance relates to the respondent's performance at varying levels of stimulus frequency (inter-stimulus intervals; ISIs), and is defined by the respondent's ability to maintain performance level even when the task rate is slow. This construct is captured by changes in the respondent's Hit Reaction Times (HRT), as indicated by the variable HRT ISI Change, as well as the observed pattern of Omissions and Commissions at various ISIs.

## Hit Reaction Time by ISI




| | 1-second ISI | 2-second ISI | 4-second ISI |
|---|---|---|---|
| HRT (ms) | 379 | 457 | 478 |
| HRT SD (ms) | 64 | 77 | 125 |

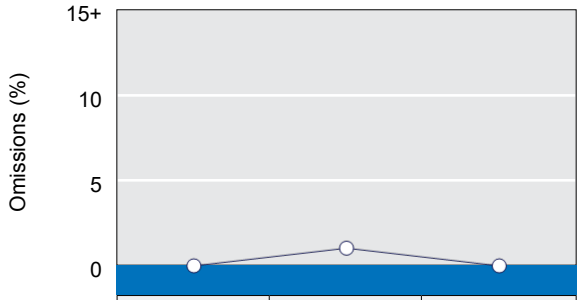
## HRT ISI Change




| | |
|---|---|
| T-score | 55 |
| Guideline | High Average |

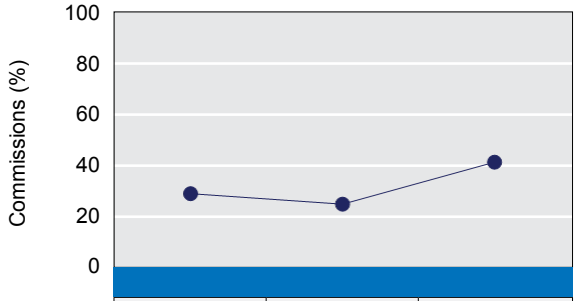
**Note.** ms = milliseconds; SD = Standard Deviation.

## Omissions by ISI




| | 1-second ISI | 2-second ISI | 4-second ISI |
|---|---|---|---|
| Omissions (%) | 0 | 1 | 0 |

## Commissions by ISI




| | 1-second ISI | 2-second ISI | 4-second ISI |
|---|---|---|---|
| Commissions (%) | 29 | 25 | 42 |

**Note**. No statistically significant differences were found in error rates between ISIs.

**HRT ISI Change** indicates the change in mean response speed at various ISIs. Kyle's T-score is 55 and falls in the **High Average** range. This result means that he had a slight reduction in response speeds at longer ISIs. There was no statistically significant increase in error rates across all three ISI levels. **Kyle's profile of scores on these measures does not indicate a problem with maintaining vigilance at varying levels of stimulus frequency.**


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## Response Style

***C*** is a signal detection statistic that measures an individual's natural response style in tasks involving a speed-versus-accuracy trade-off. Based on his or her score on this variable, a respondent can be classified as having one of the following three response styles: a *conservative* style that emphasizes accuracy over speed; a *liberal* style that emphasizes speed over accuracy; or a *balanced* style that is biased neither to speed nor accuracy. Response style can affect scores such as Commissions and Hit Reaction Time (HRT), and should be taken into consideration during interpretation.

## Detectability (*d'*)

**d-prime (*d'*)** is a measure of how well the respondent discriminates non-targets (i.e., the letter X) from targets (i.e., all other letters). This variable is also a signal detection statistic that measures the difference between the signal (targets) and noise (non-targets) distributions. In general, the greater the difference between the signal and noise distributions, the better the ability to distinguish non-targets and targets. On the Conners CPT 3, *d'* is reverse-scored so that higher raw score and *T*-score values indicate worse performance (i.e., poorer discrimination).

## Omissions (%)

**Omissions** are missed targets. High omission error rates indicate that the respondent was not responding to the target stimuli due to a specific reason (e.g., difficulty focusing). Omission errors are generally an indicator of inattentiveness.

## Commissions (%)

**Commissions** are incorrect responses to non-targets. Depending on the respondent's HRT, high commission error rates may indicate either inattentiveness or impulsivity. If high commission error rates are coupled with slow reaction times, then the respondent was likely inattentive to the stimulus type being presented and thus responded to a high rate of non-targets. If high commission error rates are combined with fast reaction times, the respondent was likely rushing to respond and failed to control his or her impulses when responding to the non-targets. In the latter case, high commission error rates would reflect impulsivity rather than inattentiveness.

## Perseverations (%)

**Perseverations** are responses that are made in less than 100 milliseconds following the presentation of a stimulus. Normal expectations of physiological ability to respond make it virtually impossible for a respondent to perceive and react to a stimulus so quickly. Perseverations are usually either slow responses to a preceding stimulus, a random response, an anticipatory response, or a repeated response without consideration of the task requirements. Perseverations may be related to impulsivity or an extremely liberal response style. Perseverations are, therefore, likely the result of anticipatory, repetitive, or impulsive responding.

## Hit Reaction Time (HRT)

**HRT** is the mean response speed, measured in milliseconds, for all non-perseverative responses made during the entire administration. An atypically slow HRT may indicate inattentiveness (especially when error rates are high), but it may also be the results of a very conservative response style. Alternatively, a very fast HRT, when combined with high commission error rates, may indicate impulsivity.

## Hit Reaction Time Standard Deviation (HRT SD)

**HRT SD** measures the consistency of response speed to targets for the entire administration. A high HRT SD indicates greater inconsistency in response speed. Response speed inconsistency is sometimes indicative of inattentiveness, suggesting that the respondent was less engaged and processed stimuli less efficiently during some parts of the administration.

## Variability

**Variability**, like HRT SD, is a measure of response speed consistency; however, Variability is a "within respondent" measure (i.e., the amount of variability the respondent showed in 18 separate sub-blocks of the administration in relation to his or her overall HRT SD score). Although Variability is a different measure than HRT SD, the two measures typically produce comparable results and are both related to inattentiveness. High response speed variability indicates that the respondent's attention and processing efficiency varied throughout the administration.

## Hit Reaction Time Block Change (HRT Block Change)

**HRT Block Change** is the slope of change in HRT across the six blocks of the administration. A positive slope indicates decelerating reaction times as the administration progressed, while a negative slope indicates accelerating reaction times. If reaction times slow down, as indicated by a higher HRT Block Change score, the respondent's information processing efficiency declines, and a loss of sustained attention is indicated.

## Omissions by Block

**Omissions by Block (raw score only)** is the rate of the respondent's missed targets in each of the six blocks. An increase in omission error rate in later blocks indicates a loss of sustained attention.

## Commissions by Block

**Commissions by Block (raw score only)** is the rate of the respondent's incorrect responses to non-targets in each of the six blocks. An increase in commission error rate in later blocks indicates a loss of sustained attention.

## Hit Reaction Time Inter-Stimulus Intervals Change (HRT ISI Change)

**HRT ISI Change** is the slope of change in reaction time across the three ISIs (1, 2, and 4 seconds). A positive slope indicates decelerating HRT at longer intervals; whereas, a negative slope indicates accelerating HRT at longer intervals. A higher HRT ISI Change score means that the respondent's information processing efficiency declined with longer pauses between stimuli, and a loss of vigilance is indicated. A significant change in response speed at the different ISIs may indicate that the respondent was having trouble adjusting to changing task demands. Sometimes, this finding relates to activation/arousal needs; some respondents may be more efficient in a busier/more stimulating environment (e.g., during the 1-second ISI) than in a less active environment where the stimuli are presented less frequently (e.g., during the 4-second ISI), or vice-versa.

## Omissions by ISI

**Omissions by ISI (raw score only)** is the rate of missed targets in each of the three ISI trial types. An increase in omission error rate on trials with longer ISIs indicates a loss of vigilance.

## Commissions by ISI

**Commissions by ISI (raw score only)** is the rate of incorrect responses to non-targets in each of the three ISI trial types. An increase in commission error rates on trials with longer ISI indicates a loss of vigilance.


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